Abdelrahman Khalifa

+20 114 250 5042 | Cairo, Egypt | abdelrahman.khalifa.gad@gmail.com | linkedin.com/in/abd002 | github.com/Abd002

Summary

Embedded software engineer writing C firmware for 32-bit microcontrollers (ESP32-S3, ARM Cortex-M STM32 and TM4C). Owns the Zephyr RTOS firmware for EKSON's pneumatic motion simulators; built a dual-bank STM32 bootloader with CRC/signature check and rollback (graduation project).

Experience

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS.
- Implemented closed-loop PID control of the pneumatic valves, with feedback from an MPU6050 IMU over I2C, plus safety stops and a watchdog.
- Cut CPU usage and latency of the PyQt5 desktop app by about 40% (measured) with asyncio.
- Built remote operation and firmware updates (Node.js/MongoDB backend, esptool) with GitHub Actions CI.

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Built the **Printers page for COSMIC Settings** as 5 Rust crates, with the CUPS/IPP/DNS-SD backend isolated behind a varlink IPC service (cosmic-settings #2182, in review).
- Wrote Rust-to-C FFI bindings (libcups3 DNS-SD), ported cups-rs to libcups3 with CI, and fixed crashes in libcups' Avahi code; 7 PRs merged upstream (PRs #18–#22, #164, #165).

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Built Linux images with Yocto and Buildroot and a kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS Mar 2025

- Implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules, merged upstream (issue #86977).

Projects

AUTOSAR-Based V2X Vehicle | *graduation project, STM32* GitHub

- Implemented the firmware update path (an ESP32/RPi gateway sends the image over UART) and a dual-bank bootloader with CRC check and rollback.

Kernel Monitor | *C, Linux, ARM, QEMU* GitHub

- Character device driver that exports per-task memory, threads, context switches and CPU time to user space using RCU.

Technical Skills

Languages: C, Modern C++, Rust, Python (asyncio, PyQt5), Java, Bash, Verilog
RTOS & firmware: Zephyr (devicetree, west), FreeRTOS, ESP-IDF, OSEK/VDX (Trampoline), 32-bit ARM Cortex-M (STM32, TM4C), ESP32-S3, AVR, AUTOSAR, bootloaders with image verification (CRC/signature), FOTA, PID control
Embedded Linux: Linux, Yocto, Buildroot, kernel modules, device drivers, device trees, QEMU, Raspberry Pi
Protocols & tools: UART, I2C, SPI, CAN, TCP/UDP, REST; Git, GitHub Actions CI, CMake, Make, debugging
Testing: unit and integration (end-to-end) testing, pytest, Python scripting for build and test automation in CI

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors.
- Activities: Codeforces Expert; 4-time ECPC finalist (2021–2024); Treasurer, IEEE Student Branch.